```r
gb <- filter(clean_transit, clean_transit$country == "UK") %>%
  mutate(country = "GB")

clean_transit<- subset(clean_transit, !e %in% c(7226, 7336, 7417))
clean_transit <- rbind(clean_transit, gb)


clean_transit$region <- countrycode(sourcevar = clean_transit$country,
                            origin = "iso2c",
                            destination = "region23")
```

```r
clean_transit <- clean_transit %>%
  mutate(group = case_when(
    region %in% c("Northern America", "Central America", "South America") ~ "Americas",
    region %in% c("Southern Europe", "Northern Europe", "Western Europe") ~ "Europe",
    region %in% c("Central Asia", "Eastern Asia") ~ "Asia",
    region %in% c("South-Eastern Asia", "Australia and New Zealand") ~ "SE Asia & Oceania",
    region %in% c("Northern Africa", "Western Asia") ~ "North Africa & West Asia",
    .default = region
  ))



#table(clean_transit$group)
df <- clean_transit
df <- df[!is.na(df$tunnel_per), ]
```

Transit Costs across the World

Data

Our transit data comes from the Transit Costs Project collaboration at the NYU Marron Institute of Urban Management. The express purpose of this dataset is to document the costs of transit projects across the world for international comparison. The transit projects under consideration all involve a railway, including light rail, trams, rapid transit/metro systems, and heavy rail (the meaning of this terminology changes based on local vernacular). This means busses, ferries, rideshare, bikes or other forms of transportation are not included in the dataset. This data was compiled by academic researchers and documents transit projects from . Each observation is a unique transit project undertaken by that country

To account for spatial factors, country shapefiles from a public ArcGIS dataset were merged into the . The map projection used in the dataset is the Mercator projection, which emphasizes accuracy of shape over size. The final dataset that was merged with the transit dataset was a cross-cultural psychology dataset from social psychologist Geert Hofstede. This dataset includes Hofstede's rankings of countries along six cultural dimensions: individualism-collectivism, uncertainty avoidance, strength of social hierarchy (called 'power-distance'), motivation toward achievement and success (formerly masculine v. feminine), and long-term vs. short term orientation. The observations in the Hofstede dataset were individual countries (or groupings of countries such as "Arab states").

Methodology

Three models were fit to the combined data.

Model 2 - Cost Prediction

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} +$$

Results

Transit Project Duration

Cost Prediction ()

Fit the baseline linear model We model **real construction cost** as a function of (i) geographic regions and (ii) key characteristics that are plausibly associated with cost: project year, length of line, number of stations, purchasing power parity (PPP), and construction duration.

$$real_cost \sim region + year + length + stations + ppp_rate + duration$$

The initial model statistics is

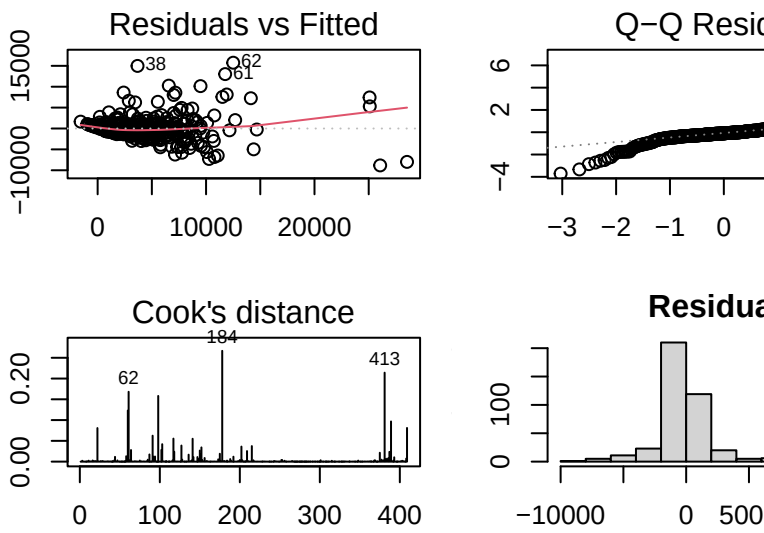
```
full <- lm(real_cost ~ group + year+ length + stations+ ppp_rate+duration, data = df)

broom::glance(full) %>%
  dplyr::select(r.squared, sigma, AIC, BIC, nobs) %>%
  knitr::kable(digits = 3, caption = "Model fit statistics")
```

Table 1: Model fit statistics

r.squared	sigma	AIC	BIC	nobs
0.681	2592.439	7604.282	7656.46	409

Linear regression relies on zero means, constant error variance, and approximately normal residuals. We evaluate these using standard residual diagnostics. In the baseline model, residual plots indicate heteroskedasticity and departures from normality.

```
par(mfrow = c(2,2), mar=c(3,3,2,1))
plot(full, which = 1)
plot(full, which = 2)
plot(full, which = 4)
hist(resid(full), main = "Residuals", xlab = "")
```



Diagnostic assessment of modeling assumptions

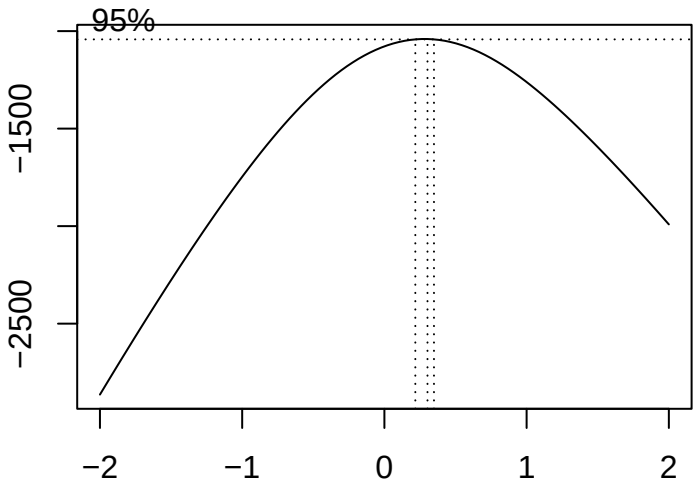
```
par(mfrow = c(1,1))
```

Transformation via BoxCox To address non-constant variance and improve residual normality, we apply a Box–Cox transformation to the strictly positive response.

```
library(MASS)
```

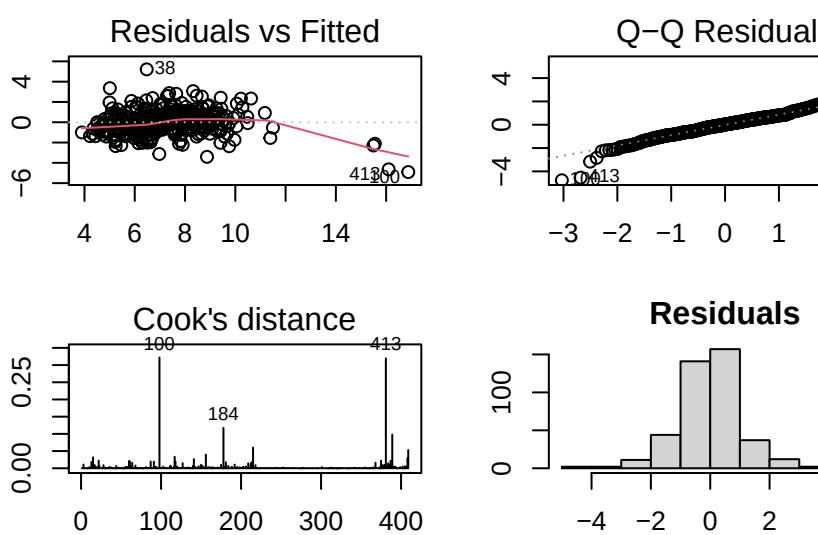
```
##
## Attaching package: 'MASS'

## The following object is masked from 'package:dplyr':
##
##     select
```

```
df_pos <- df %>% filter(real_cost > 0)
full_pos <- lm(real_cost ~ group + year + length + stations + ppp_rate + duration, data = df_pos)
par(mar = c(3,3,2,1))
MASS::boxcox(full_pos, plotit = TRUE) ### The result turns out that we should raise y to power of 0.25
```



The Box–Cox log-likelihood plot suggests a transformation near $\lambda \approx 0.25$, so we proceed with $Y^{0.25}$

```
full_trans <- lm(real_cost^0.25 ~ group + year+ length + stations+ ppp_rate+duration, data = df)
#summary(full_trans)

par(mfrow = c(2,2), mar=c(3,3,2,1))
plot(full_trans, which = 1)
plot(full_trans, which = 2)
plot(full_trans, which = 4)
hist(resid(full_trans), main = "Residuals", xlab = "")
```



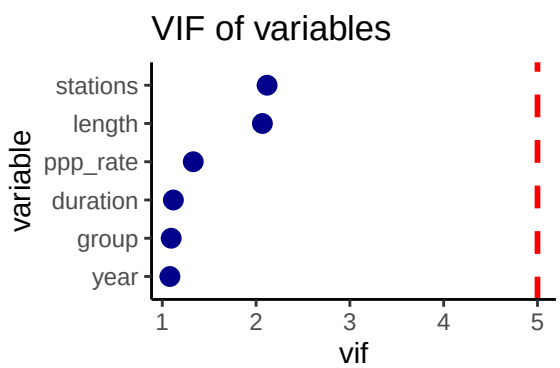
Refit our model and check conditions again.

```
par(mfrow = c(1,1))
```

After transformation, the residual diagnostics improve noticeably. The TA plot suggests more constant variance, and the Q–Q plot shows residuals closer to normality.

We also examine influential observations. Although some points have larger standardized residuals, influence depends on both residual size and leverage, summarized by Cook's distance. All Cook's distances are well below common thresholds, 0.5, so no observations are removed.

```
v <- vif(full_trans)

vif_df <- data.frame(
  variable = rownames(v),
  vif = v[,3]
)


ggplot(vif_df, aes(x = vif, y = reorder(variable, vif)))+
  geom_point(size = 3, color = 'darkblue')+
  geom_vline(xintercept = 5, linetype = 'dashed', color = 'red', linewidth=1 )+
  labs(title = 'VIF of variables', y ='variable')+
  theme_classic()
```



Multicollinearity assessment

We assess multicollinearity using variance inflation factors (VIF). Values exceeding 5 are often used as thresholds to indicate high collinearity. The VIF results here do not indicate problematic collinearity.

Model selection

Backward model selection To assess whether a simpler model provides similar explanatory power, we apply backward model selection using AIC starting from the transformed-response model. Because AIC balances model fit and complexity, smaller values indicate a better model.

```
step_fit <- step(full_trans,
     direction = 'backward', trace  = 0)

data.frame(
  Initial_AIC = round(AIC(full_trans),2),
  End_AIC = round(AIC(step_fit),2),
  Equation = paste(deparse(formula(step_fit)), collapse = ''))%>%
knitr::kable(caption = "Backward AIC selection summary")
```

Table 2: Backward AIC selection summary

Initial_AIC	End_AIC	Equation
1263.57	1263.57	real_cost^0.25 ~ group + year + length + stations + ppp_rate + duration

Selection result. The backward AIC procedure does not remove any predictors, indicating that each variable contributes meaningfully to model fit under the AIC criterion. It indicates that removing any single predictor increases AIC relative to the full transformed model.

Final Model

We proceed with the transformed-response model including all pre-specified predictors:

This model achieves improved adherence to the regression assumptions after the transformation, including approximately constant variance and residuals that are closer to normally distributed. Retaining all predictors also preserves interpretability with respect to the study objective, allowing us to evaluate how project characteristics and geographic region jointly influence transit construction costs. Below is the full summary of the model.

```
# coefficient table
broom::tidy(full_trans) %>%
  dplyr::select(term, estimate, std.error, p.value) %>%
  knitr::kable(digits = 3,caption = "Summary of the transformed regression model.")
```

Table 3: Summary of the transformed regression model.

term	estimate	std.error	p.value
(Intercept)	-87.902	20.978	0.000
groupAsia	-0.503	0.239	0.036
groupEastern Europe	-0.897	0.328	0.007
groupEurope	-1.702	0.248	0.000
groupNorth Africa & West Asia	-0.816	0.294	0.006
groupSE Asia & Oceania	0.638	0.310	0.040
groupSouthern Asia	-0.813	0.315	0.010
year	0.047	0.010	0.000
length	0.049	0.006	0.000
stations	0.027	0.008	0.001
ppp_rate	-0.149	0.082	0.068
duration	0.069	0.020	0.001

```
# model fit
broom::glance(full_trans) %>%
  dplyr::select(r.squared, adj.r.squared, sigma) %>%
  knitr::kable(digit = 3)
```

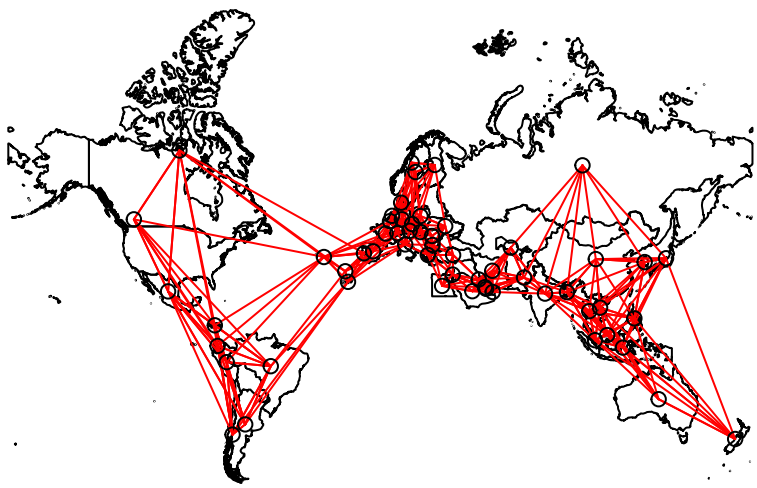
r.squared	adj.r.squared	sigma
0.694	0.686	1.115

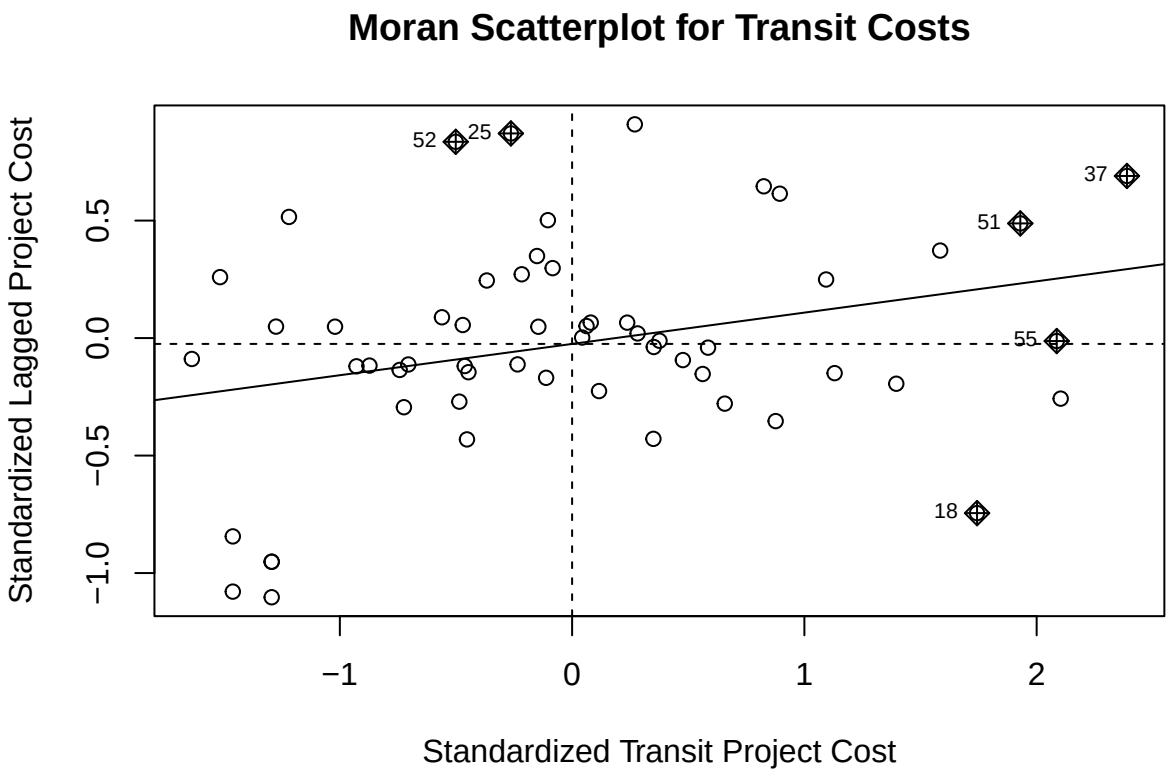
Spatial and Cultural Predictors (Kevin Ji)

A particular concern for our prior models is spatial autocorrelation. There is strong reason to believe that the presence of transit in neighboring countries could disincentivise or incentivise countries to build it themselves. If a country's passenger rail buildout is successful, it encourages neighbors to invest in rail projects themselves. Certain countries have legal frameworks or cultural biases that discourage infrastructure investments like transit, and these features are often shared among neighboring countries. Spatial autocorrelation may also capture the aspects of geography that make it more costly or inexpensive to build in certain regions, such as soil quality or climate.

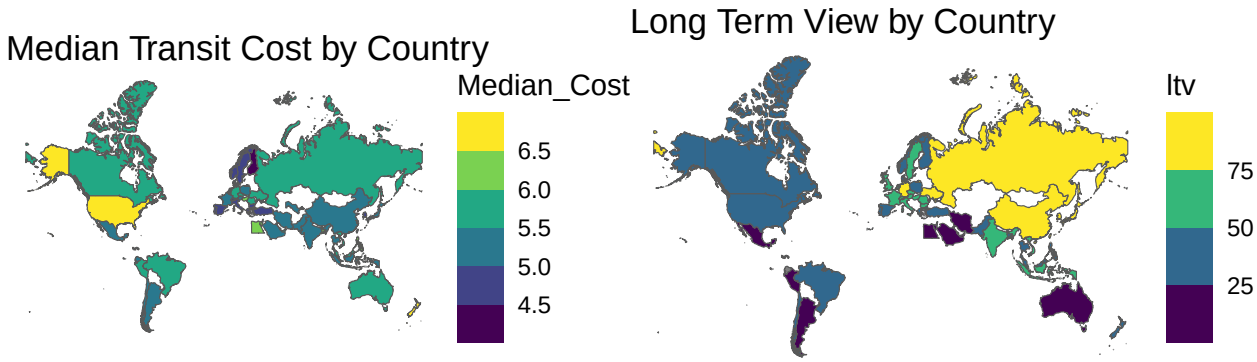
To model spatial autocorrelation, we import shapefiles from the

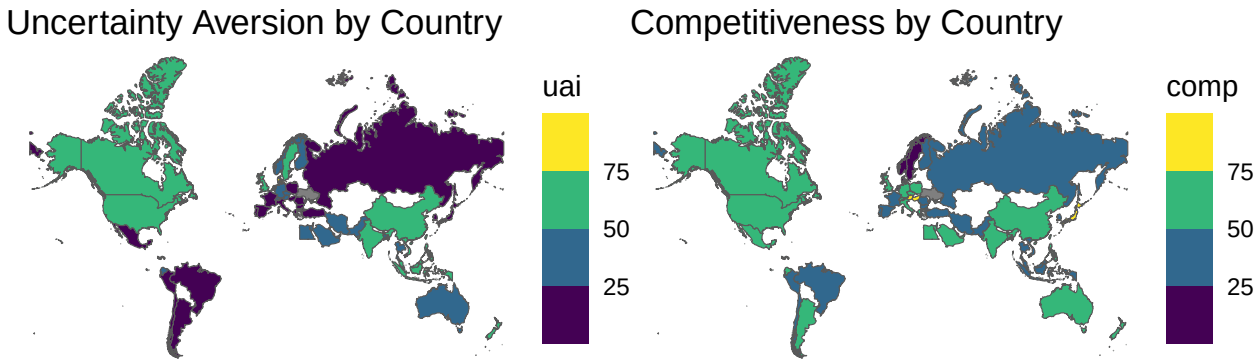
```
plot(st_geometry(medshp))
plot(dist_nb, coords = st_coordinates(centroid), add = TRUE, col = "red")
```






```
## Warning: There were 5 warnings in ‘mutate()‘.
## The first warning was:
## i In argument: ‘longtermview = as.numeric(ltowvs)‘.
## Caused by warning:
## ! NAs introduced by coercion
## i Run ‘dplyr::last_dplyr_warnings()‘ to see the 4 remaining warnings.
```





Some econometric procedures recommending accounting for spatial autocorrelation by inflating standard errors.

```
##                     GVIF Df GVIF^(1/(2*Df))
## uncertainty     2.379687  1        1.542623
## power_distance  2.232496  1        1.494154
## competitive     1.535409  1        1.239116
## year            1.226286  1        1.107378
## length          1.305438  1        1.142558
## ppp_rate        2.862822  1        1.691987
## rr              1.225111  1        1.106847
## tunnel_per      1.465959  1        1.210768
## duration        1.466593  1        1.211030
## group          14.763651  6        1.251506
```

```
## Non-constant Variance Score Test
## Variance formula: ~ fitted.values
## Chisquare = 0.03064366, Df = 1, p = 0.86104
```

Conclusion